

Abdullah Mahmoud Fawzy

Frontend / Fullstack Engineer

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Professional Summary

Software Engineer with a foundational degree in Mathematics from Helwan University, specializing in modern Frontend and Fullstack engineering using Next.js 16 (App Router), React 19, TypeScript, and Node.js. Experienced in designing maintainable web architectures using SOLID principles and Strategy/Factory patterns, implementing in-memory rate limiting, securing data flows with HTTP-only cookies and PostgreSQL RLS, and writing automated unit tests with Jest. Passionate about mathematical problem solving, clean code standards, and production-grade reliability.

Education

Bachelor of Science in Mathematics, *Helwan University* 10/2024
Cairo, Egypt

Technical Skills

Languages

TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

UI & Styling

Tailwind CSS v4, Radix UI Primitives, CSS Modules, Responsive Design, Accessible UI (ARIA), Recharts

Databases & Cloud

PostgreSQL, Supabase, MongoDB, Mongoose, Vercel, Git & GitHub

Frontend

Next.js 16 (App Router, Server & Client Components, Server Actions), React 19, React Hooks, Redux Toolkit, TanStack Query

Backend & APIs

Node.js, Express.js, RESTful APIs, WebSockets (Socket.IO), JWT Authentication, Zod Validation, Rate Limiting

Architecture & Engineering

Automated Unit Testing (Jest), SOLID Principles, Strategy Pattern, Factory Pattern, Server-Side Prefetching & Cache Hydration

Projects

Devfolio AI — Engineering Showcase & Architectural Assistant,

Next.js 16, React 19, TypeScript, Tailwind CSS v4, Jest, MongoDB

- Architected an AI gateway decoupling LLM providers via Strategy and Factory patterns, supporting seamless provider swapping and 100% deterministic Jest unit testing.
- Built an in-memory sliding-window rate limiter and a custom LLError hierarchy to protect API quotas and propagate exact HTTP status codes (429, 503, 500).

Inventory Management & Analytics Dashboard, *Next.js 16, TypeScript, TanStack Query, Tailwind CSS, Recharts*

- Implemented server-side data prefetching with client-side cache hydration using TanStack Query, eliminating layout shifts and redundant network requests.
- Built multi-parameter filtering, debounced search, and pagination across 1,000+ inventory records paired with interactive Recharts financial analytics.

Real-Time Team Chat & Collaboration Platform, *React, Redux Toolkit, Node.js, Express, MongoDB, Socket.IO, JWT*

- Engineered bidirectional messaging using Socket.IO 📠 room broadcasting with optimistic UI updates and normalized Redux Toolkit state management.
- Implemented stateless JWT authentication using secure HTTP-only cookies and protected Express middlewares to prevent token exfiltration.

Hotel Management & Operations Dashboard, *React, Supabase (PostgreSQL), React Query, React Hook Form, Recharts*

- Integrated Supabase PostgreSQL with declarative Row Level Security (RLS) policies to enforce database-level access control.
- Built end-to-end booking reservation workflows with strict React Hook Form validation, modal flows, and occupancy rate analytics.

Languages

Arabic
Native

English
Professional Working Proficiency